



Bilgecag Aydogdu, PhD

Data Scientist | Mobile Phone Data Specialist

Email: bilgecagaydogdu@gmail.com

Based: Paris, France

Nationality: Dutch (EU citizen)

Professional Profile

Applied Data Scientist with 5+ years of experience specializing in **mobile phone data for official statistics and migration analytics**. Proven track record of processing large-scale telecommunications datasets (CDR/xDR), developing reproducible methodologies for mobility analysis, and collaborating with national statistical offices and UN agencies. Strong background in Python, SQL, statistical modeling, and communicating insights to non-technical stakeholders. Published 4 peer-reviewed papers on mobile phone data methodologies.

Relevant Work Experience

Mar 2026– **Data Scientist, Positium, Estonia (Remote).**

- Delivered interactive geospatial dashboards in Python and JavaScript, enabling non-technical stakeholders to self-serve.
- Built production-scale data pipelines in Apache Spark that transform billions of daily mobile network events from millions of subscribers into population and mobility statistics, aggregated by spatial grid and time interval.
- Researched, designed, and implemented a present-population estimation component, evaluating competing methodological approaches.
- Developed validation frameworks for origin-destination mobility matrices, using gravity models, intraday movement patterns, and commute directionality checks to verify outputs against real-world expectations

Mar 2020–Jan 2026 **PhD Researcher – Mobile Phone Data for Migration Statistics, Utrecht University, Netherlands.**

- Developed novel methodology for measuring population displacement using mobile positioning data, now published in peer-reviewed journal (EPJ Data Science)
- Processed and analyzed CDR and xDR data from millions of users to produce migration statistics
- Built reproducible data pipelines in Python for processing millions of daily mobile phone events
- Integrated multiple data sources: telecommunications data, satellite imagery, survey data, administrative boundaries
- Applied statistical methods for debiasing and validating mobility estimates
- Collaborated with national statistical offices and international organizations on methodology development
- Delivered training workshops on mobile phone data analysis to government statisticians

2020–2024 **Data Science Researcher, Turkcell (Major Telecom Operator), Istanbul, Turkey (Remote).**

- Engineered complex SQL queries in Hadoop environment to process large-scale mobile positioning datasets
- Built ETL pipelines for processing millions of subscriber transactions and behavioral data points daily
- Developed data quality control processes, anomaly detection systems, and automated monitoring
- Collaborated with cross-functional teams on data infrastructure, governance, and reporting solutions
- Worked with distributed databases for large-scale data management

Jun 2022 **Consultant – Mobile Phone Data Training, International Organization for Migration (IOM), Mongolia.**

- Designed and delivered capacity-building workshop for Mongolian National Statistical Office
- Trained government statisticians on integrating mobile phone data with traditional census methods

Education

2020–2025 **PhD in Computer Science, Utrecht University, Netherlands.**

Thesis: Mobile Phone Data for Crisis-Induced Migration and Mobility

2017–2019 **M.Phil. Development Economics, Paris 1 Panthéon-Sorbonne University, France, Bien (Honours).**

Advanced coursework in Statistics, Econometrics, Causal Inference

2013–2017 **B.A. Economics and Political Science, Bogazici University, Istanbul, Turkey, High Honours.**

Selected Publications

- 2025 **Aydoğdu, B., et al.** “A novel activity space approach to discover displacement patterns via mobile phone data: An analysis of the 2023 Türkiye-Syria Earthquakes.” *EPJ Data Science*. [Open-source code available]
- 2025 **Aydoğdu, B., et al.** “Mobile phone data for anticipating displacements: practices, opportunities, and challenges.” *Data & Policy*.
- 2025 **Aydoğdu, B., et al.** “Analyzing international airline top-up transfers for migration and mobility.” *International Journal of Data Science and Analytics*.

2025 Arcila-Calderón, C., **Aydoğdu, B.**, et al. "Combining Twitter and mobile phone data to observe border-rush." *Journal of Computational Social Science*.

Technical Skills

Programming Python (pandas, PySpark, scikit-learn, numpy, scipy, geopandas), R, MATLAB
Data & Tools SQL (PostgreSQL, DuckDB), Spark, Hadoop/Hive, Airflow, Git, Docker
Big Data Hadoop ecosystem, large-scale data processing pipelines

Languages

English Fluent (C2)
French Professional (C1)
Turkish Native
Dutch Basic (A2)

Awards & Grants

2020–2024 Horizon 2020 Project Grant: HumMingBird project funded by European Commission
2017–2019 Eiffel Excellence Scholarship: French Ministry for Europe and Foreign Affairs